

SignNav: Directional Signage-Cue Guided Visual Navigation in Large-Scale Indoor Environments

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Abstract—Humans routinely leverage directional cues from signage to navigate toward destinations in unfamiliar Large-Scale Indoor (LSI) environments, such as hospitals and airport terminals. However, learning how embodied agents can use sparse and target-relevant signage cues for closed-loop navigation remains underexplored. This paper studies SignNav, a directional signage-cue guided visual navigation task in which an agent navigates to a semantic target using egocentric observations together with directional cues and cue locations provided by an upstream sign parser. This formulation focuses on downstream cue-conditioned policy learning, rather than open-ended signage parsing, OCR, or compound sign reasoning. To facilitate research in this domain, we construct the LSI-Dataset, a large-scale benchmark with signage cues-labeled indoor environments and automatically generated trajectory-action episodes. SignNav presents two key challenges: grounding transient directional cues into the current observation, and maintaining temporal memory when signage is sparse or temporarily unavailable. To address these challenges, we propose the Spatial-Temporal Aware Transformer (START), which combines a spatial-aware module for cue-observation grounding with a temporal-aware module for history-based decision making. Extensive experiments, including cue-representation ablations, sparse-signage evaluations, and real-world robot trials, demonstrate the effectiveness of START for directional signage-cue guided navigation in large-scale indoor environments.

Index Terms—Autonomous agents, perception-action coupling, signage-driven navigation, vision-based navigation.

I. INTRODUCTION

NAVIGATING Large-Scale Indoor (LSI) environments, such as airport terminals and hospitals, remains a fundamental challenge for embodied agents. In such spaces, target

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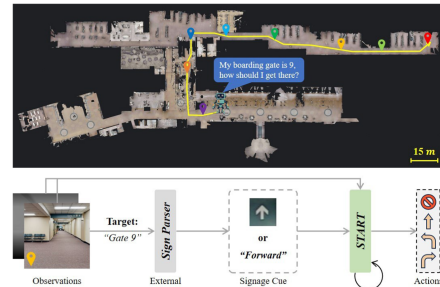


Fig. 1. Directional signage-cue guided navigation in a LSI environment. Humans use signage to extract target-relevant directional cues when navigating unfamiliar public spaces. SignNav studies the downstream navigation problem in which an agent uses parsed directional cues from signage, together with egocentric observations and history, to reach the target location. The directional signage-cue can be represented as either visual crop image or structured category.



Fig. 2. An episode example in a hospital scene. The agent is required to make action decisions according to the directional signage-cue and finally arrive at the target location. Different scenes have different styles of signage.

locations are often not directly visible from the start, and successful navigation requires the agent to exploit intermediate guidance provided by the environment. Humans naturally use signage for this purpose: rather than following a pre-given step-by-step instruction, they usually observe local signs, extract target-relevant directional cues, and use these cues to decide how to proceed (see Fig. 1). Enabling robots to use such environment-embedded guidance is therefore an important step toward practical indoor navigation.

Existing embodied navigation paradigms [1] have made substantial progress but do not fully capture this setting. Object Navigation (ObjectNav) [2] typically requires the agent to search for an object category and assumes that the goal can eventually be visually recognized, while Vision-and-Language Navigation (VLN) [3] relies on a complete natural-language instruction given before navigation starts. Recent methods based on large

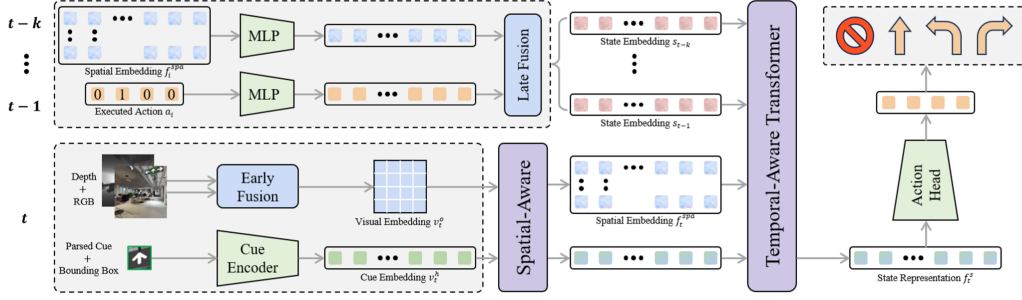


Fig. 3. The architecture of our Spatial-Temporal Aware Transformer (START) model for the SignNav task. START uses a spatial-aware module to ground the parsed cue and its location into the current observation, and a temporal-aware module to integrate historical states when cues are sparse or temporarily unavailable. The parsed cue can be represented as either visual crop image or structured category, such as “forward” and “right”.

language or vision-language models improve open-vocabulary reasoning [4], [5], [6], [7], but they are often evaluated in relatively small household environments or depend on modular pipelines with explicit mapping and planning. In contrast, signage-guided navigation in LSI environments requires the agent to react to sparse and dynamically observed cues that become available only when the agent reaches the corresponding regions.

In this work, we study **SignNav**, a directional signage-cue guided visual navigation task for large-scale indoor environments. To clarify the scope of the problem, we separate sign perception from downstream navigation. Given a target g , an upstream sign parser extracts the target-relevant directional cue and its location from the current observation. The navigation policy then uses this parsed cue, together with egocentric observations and navigation history, to predict the next action. This formulation focuses on how an embodied agent grounds directional signage cues into physical navigation behavior, rather than claiming to solve open-ended signage parsing, compound sign reasoning, or general visual sign grounding.

To support this task, we construct the **LSI-Dataset**, which contains signage-labeled LSI environments and automatically generated trajectory-action episodes. The dataset is built through a procedural signage injection and annotation pipeline, allowing directional cues and their spatial locations to be recorded during navigation. SignNav introduces two key challenges for robots. (1) **Dynamic Cue Grounding**: The same cue can imply different actions depending on where it appears in the egocentric scene and what navigable structure is currently visible. (2) **Sparse Observability**: Signage is sparse and may be observed before the corresponding junction becomes actionable, requiring temporal memory to retain previously observed cues and use them later. These challenges distinguish **SignNav** from settings where the instruction is fixed at the beginning of the episode or where the goal object is directly searched for in the scene.

To address these challenges, we propose the **Spatial-Temporal AwaRe Transformer (START)**, a closed-loop policy for directional signage-cue guided navigation (Fig. 3). START uses a spatial-aware module to ground the parsed cue and its location into the current egocentric observation, and a temporal-aware module to integrate historical states when cues are sparse or temporarily unavailable. We evaluate START against representative rule-based, VLM-based, and learning-based baselines, and

further analyze cue representations, bounding-box information, signage sparsity, and real-world deployment. The results show that START effectively exploits directional signage cues for navigation in large-scale indoor environments, while also revealing important limitations and future directions for richer signage understanding and more complex public-space behaviors.

II. RELATED WORK

A. Object Goal Navigation

Object Goal Navigation (ObjectNav) [2] requires an agent to find an instance of a specified object category in an unseen environment. Existing methods commonly rely on reinforcement learning [8], [9], imitation learning [10], semantic mapping [11], [12], or open-vocabulary visual representations [4], [5], [13], [14] to improve search efficiency. However, ObjectNav is mainly designed for targets that can eventually be visually recognized. In LSI environments, destinations such as boarding gates or service areas may not be directly observable from a distance, and navigation often depends on intermediate directional information provided by signage. Our work therefore focuses on using target-relevant directional cues encountered during navigation rather than directly searching for the final target object.

B. Vision-and-Language Navigation

Vision-and-Language Navigation (VLN) studies how agents follow natural-language instructions to reach a goal [3], [15]. Recent extensions use pre-trained vision-language models or large language models to improve instruction following and generalization [7], [16], [17], [18]. Signage-guided navigation is closely related to VLN because signs can be regarded as environment-embedded instructions. The difference is that signage cues are local, sparse, and observed online during execution, rather than being provided as a complete global instruction at the beginning of an episode. SignNav studies this setting by conditioning navigation on target-relevant directional cues extracted from visible signage.

C. Signage-Aware and Cue-Conditioned Navigation

Signage has been studied in robotics and computer vision through sign detection [19], [20], text parsing [21], [22], visual sign grounding [23], [24], and mapless navigation [25]. Recent

works such as ReasonNav [26] and SignScene [27] further highlight the importance and difficulty of interpreting navigational signs in real environments, including the grounding of symbolic sign information to local spatial structure. Similarly, concurrent works [28], [29] explore signage-based navigation but rely on modular pipelines that pair LLMs with traditional planners. Other cue- or command-conditioned navigation methods, such as DECISION [30] and Intention-Net [31], learn navigation behaviors from discrete intentions or local goal cues. These works are highly relevant to our setting, but they either address broader sign understanding and modular reasoning, or assume that high-level commands are already available for low-level control. In contrast, our work focuses on downstream directional-cue guided visual navigation: given a parsed target-relevant cue and its location, the agent learns to ground this information in egocentric observations and leverage temporal context for predicting closed-loop navigation actions.

III. DIRECTIONAL SIGNAGE-CUE GUIDED VISUAL NAVIGATION

This work focuses on downstream navigation conditioned on signage-derived directional cues in large-scale indoor environments. We decouple raw signage parsing from policy learning: given the current observation and target, an upstream parser (such as [22]) provides the target-relevant directional cue and its image location, while the navigation policy grounds this cue in the egocentric observation and incorporates temporal context to predict actions. This formulation isolates directional-cue guided navigation from open-ended OCR, compound sign reasoning, and general signage understanding. In simulation, the parser is implemented by procedural annotations; in real-world deployment, it can be instantiated by a vision-language sign parser.

A. Task Definition

We formulate **SignNav** as a Partially Observable Markov Decision Process (POMDP). The objective is for an embodied agent to navigate in a continuous 3D environment toward a semantic target g by using target-relevant directional cues observed from signage.

Observation Space: At each time step t , the agent receives an egocentric RGB-D observation $o_t = \{o_t^r, o_t^d\}$. An upstream sign parser extracts the directional cue and its image location as:

$$\psi(o_t, g) \rightarrow (c_t, b_t) \quad (1)$$

where c_t denotes the parsed target-relevant cue and $b_t = [x_{min}, y_{min}, x_{max}, y_{max}]$ denotes its bounding box. The cue c_t can be represented as either a visual crop or a structured direction category, depending on the parser interface.

Action Space: The agent uses a discrete action space $a_t \in \mathcal{A}$ consisting of four low-level commands: Move Forward (0.25m), Turn Left (15°), Turn Right (15°), and Stop. This setup follows common continuous-environment navigation protocols [15].

Policy Learning: Since signage cues are sparse and may be observed before the corresponding junction becomes actionable, the agent must use historical context. We define the history as:

$$\mathcal{H}_t \triangleq \{(o_0, c_0, b_0, a_0), \dots, (o_{t-1}, c_{t-1}, b_{t-1}, a_{t-1})\} \quad (2)$$

The goal is to learn a policy π , parameterized by Θ , that predicts the next navigation action according to:

$$a_t \sim \pi(a_t | \mathcal{H}_t, o_t, c_t, b_t; \Theta) \quad (3)$$

The episode terminates when the agent issues Stop near the target or reaches the maximum step limit.

B. LSI-Dataset Creation

Existing embodied navigation datasets [32], [33] mainly focus on household-scale environments and do not fully capture the spatial scale and topology of public indoor spaces. To support **SignNav**, we construct the **Large-Scale Indoor (LSI) Dataset** from 20 high-fidelity 3D scans collected from open-source repositories [32], [34], [35], including hospitals, offices, and other large indoor environments. These scenes contain long corridors, open halls, and complex topological structures, making them suitable for evaluating long-horizon signage-cue guided navigation.

Signage Injection: We develop an automated Blender-based pipeline to procedurally inject signage into the 3D meshes. Candidate sign locations are selected around decision points such as intersections and corners, and directional cues are assigned according to the local route geometry. To reduce overfitting to a single visual template, the dataset includes diverse signage appearances, including variations in arrow shape, color, layout, and background style. Fig. 2 illustrates the rendered effect in a hospital scene. In this work, we focus on directional cues and leave richer signage understanding, such as OCR and compound instruction parsing, to upstream perception modules.

Cue Representation: For each visible target-relevant sign, the dataset provides the directional cue in two complementary forms: (1) a visual cue, represented by the cropped image region of the directional symbol, and (2) a structured cue, represented by the parsed direction category, e.g., *left*, or *forward*. Both cue forms are paired with the same image-space bounding box b_t , allowing models to study either visual cue grounding or parser-based structured cue grounding under a unified navigation protocol.

Episode Generation: For each scene, we construct a navigable connectivity graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ using Poisson Disk Sampling [36]. Start-goal pairs are sampled from graph nodes $\mathcal{V}$, and shortest paths are planned using Dijkstra's algorithm followed by spline-based smoothing. An oracle agent then executes the smoothed trajectory to generate training and evaluation episodes. To automatically annotate the directional signage cues for each episode, we determine the oracle cue direction by leveraging the local curvature of the smoothed trajectory and the agent's heading relative to the injected signage. At each time step t , we record the RGB-D observation o_t , the target-relevant directional cue c_t , its bounding box b_t , and the oracle action a_t^* :

$$\tau = \{(o_t, c_t, b_t, a_t^*)\}_{t=0}^T \quad (4)$$

where T is the episode horizon. In simulation, c_t and b_t are obtained from procedural annotations, providing a controlled interface for studying downstream directional-cue guided navigation.

IV. SPATIAL-TEMPORAL AWARE TRANSFORMER

Fig. 3 illustrates the architecture of the proposed START model for directional signage-cue guided navigation. At each time step t , START receives the egocentric RGB-D observation o_t , a target-relevant directional cue c_t , and its bounding box b_t from the upstream parser. The cue can be represented either as a visual crop or as a structured direction category. START first encodes the observation and cue, then uses a *Spatial-Aware Transformer* to ground the cue in the current egocentric scene. To handle sparse or delayed signage observations, a *Temporal-Aware Transformer* further aggregates historical states and actions. The resulting state representation is finally passed to an action head to predict the next discrete navigation action.

A. Input Embeddings

Observation Embedding: Given the RGB-D observation $o_t = \{o_t^r, o_t^d\}$, we encode the RGB image and depth image using two lightweight CNN encoders with the same architecture but different input channels. The resulting features, v_t^r and v_t^d , are fused and normalized to obtain the observation embedding:

$$v_t^o = \text{LayerNorm}(v_t^r + v_t^d) \quad (5)$$

This embedding provides both appearance and geometric information for subsequent cue grounding.

Cue Embedding: The target-relevant directional cue c_t can be represented in two forms: a visual cue crop or a structured direction category. For the visual-cue interface, we encode the cropped cue image using a ResNet-50 [37] pre-trained on ImageNet. For the structured-cue interface, we encode the parsed direction category with a learnable category embedding. In both cases, the cue is paired with its image-space bounding box b_t . We normalize the box coordinates as $f_t^b = [\frac{x_{\min}}{W}, \frac{y_{\min}}{H}, \frac{x_{\max}}{W}, \frac{y_{\max}}{H}, \frac{w \cdot h}{W \cdot H}]$, where W and H are the image width and height. The final cue embedding is obtained by projecting and fusing the cue feature f_t^c and the box feature f_t^b as:

$$v_t^c = \text{LayerNorm}(f_t^c W^c) + \text{LayerNorm}(f_t^b W^b) \quad (6)$$

where W^c and W^b are learnable projection matrices. This design allows START to use either visual cue grounding or parser-based structured cue grounding under the same spatial-temporal policy architecture.

B. Spatial-Aware Transformer

Fig. 4 illustrates the navigation action decisions made by the agent leveraging directional signage-cue under different circumstances. The same directional cue may lead to different actions depending on the agent's pose, cue location, and visible scene structure. Conversely, different cues observed at the same location may require different navigation decisions. Therefore, START must ground the target-relevant cue in the current egocentric observation rather than directly treating the cue as an executable action command.

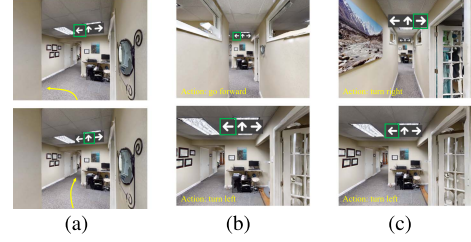


Fig. 4. **Different action decisions under different circumstances.** (a) different directional cues at the same location; (b) the same directional cues at different locations; (c) different directional cues at different locations.



Fig. 5. **One period in an episode that there is no visible directional cue of signage.** The agent still needs to make correct decisions by incorporating historical states.

Given the observation embedding v_t^o and cue embedding v_t^c , we use a *Spatial-Aware Transformer* to model their interaction inspired by the Vision Transformer (ViT) [38]. We first reshape v_t^o into a sequence of flattened visual patches $v_t^{patch} \in \mathbb{R}^{N \times (P^2 \cdot C_o)}$, where P is the patch size and N is the number of patches. These patches are projected into a D -dimensional latent space. Instead of initializing the [CLS] token independently, we initialize it with the projected cue embedding v_t^c , so that the cue serves as the query for retrieving relevant visual regions. The self-attention blocks in *Spatial-Aware Transformer* is computed as follows:

$$Q = X_{l-1} W_{l,k}^Q, K = X_{l-1} W_{l,k}^K, V = X_{l-1} W_{l,k}^V \quad (7)$$

$$H_{l,k} = \text{Softmax} \left(\frac{QK^\top}{\sqrt{d_h}} \right) V \quad (8)$$

where W^Q , W^K , and W^V are learnable projections, and d_h is the attention head dimension. The resulting patch features form the spatial embedding f_t^{spa} , while the updated cue token summarizes the cue-conditioned scene context for subsequent temporal reasoning. This module is shared by both visual-cue and structured-cue interfaces.

C. Temporal-Aware Transformer

In LSI environments, directional signage cues are not continuously visible. A cue may disappear because of the limited field of view, or it may be observed before the corresponding junction becomes actionable (as shown in Fig. 5). Therefore, the agent must retain previously observed cue-grounded information and combine it with the current observation. To this end, START uses a *Temporal-Aware Transformer* to model dependencies across recent navigation states.



Fig. 6. **Trajectories visualization of our method compared to baselines.** The second row shows the first-person-view of agent as it navigates along the ground-truth trajectory.

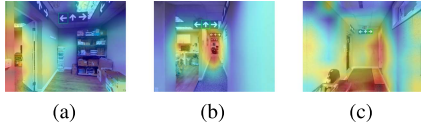


Fig. 7. **Attention map visualization both with the spatial-aware and temporal-aware modules in our START model.** The darker the color, the greater the attention weight.

Let $\mathcal{H}_k = \{(f_{t-k}^{spa}, a_{t-k}), \dots, (f_{t-1}^{spa}, a_{t-1})\}$ denote a fixed-length history buffer, where f_i^{spa} is the spatial embedding produced by the Spatial-Aware Transformer and a_i is the executed action. For each historical step i , we aggregate f_i^{spa} by average pooling and fuse it with the action embedding:

$$s_i = \text{LayerNorm}(e_i^{spa} \mathbf{W}^{spa}) + \text{LayerNorm}(e_i^a \mathbf{W}^a) \quad (9)$$

where e_i^{spa} denotes the pooled spatial feature and e_i^a denotes the embedded action.

The sequence of historical states $\{s_{t-k}, \dots, s_{t-1}\}$, together with the current cue-conditioned state s_t , are then processed by *Temporal-Aware Transformer*. The k -th attention head at the l -th layer performs self-attention over $\mathbf{X}_{l-1}$ following (7) and (8). The outputs from all the attention heads will be concatenated and projected onto the same dimension as the input as:

$$\mathbf{H}_l = [\mathbf{H}_{l,1}; \dots; \mathbf{H}_{l,k}] \mathbf{W}_l^o \quad (10)$$

where k is the total number of attention heads, $[\cdot]$ denotes concatenation. Finally, the output of layer l is formulated by:

$$\mathbf{H}'_l = \text{LayerNorm}(\mathbf{H}_l + \mathbf{X}_{l-1}) \quad (11)$$

$$\mathbf{X}'_l = \text{GeLU}(\mathbf{H}'_l \mathbf{W}_l^{F_1}) \mathbf{W}_l^{F_2} \quad (12)$$

$$\mathbf{X}_l = \text{LayerNorm}(\mathbf{H}'_l + \mathbf{X}'_l) \quad (13)$$

where GeLU is the Gaussian Error Linear Unit activation function. The final state representation f_t^s encodes both the current cue-grounded observation and recent navigation history, enabling the policy to act when signage is temporarily unavailable or when a previously observed cue must be executed later.

D. Training Objective

We train START as a closed-loop action prediction policy for directional signage-cue guided navigation. Given the state representation f_t^s produced by the spatial-temporal modules, an MLP action head predicts the probability distribution over the discrete action space:

$$p_t^a = \text{Softmax}(f_t^s \mathbf{W}^s) \quad (14)$$

where $\mathbf{W}^s$ denotes learnable parameters. The policy is first optimized by imitation learning from oracle trajectories. We minimize the inflection-weighted cross-entropy loss:

$$\mathcal{L}_{loss} = - \sum_t w_t a_t^* \log(p_t^a) \quad (15)$$

where a_t^* is the oracle action and w_t is the inflection weight, which emphasizes timesteps where the expert action changes.

To reduce covariate shift during closed-loop execution, we further fine-tune the policy using Dataset Aggregation (Dagger) [39]. At each iteration, the current policy interacts with the environment, and the resulting states are relabeled by the oracle and added to the training set. This two-stage training strategy improves robustness when the agent deviates from the reference trajectory.

V. EXPERIMENTS

A. Experimental Setup

Dataset: We train and evaluate all methods in Habitat on the LSI-Dataset. The dataset contains 20 LSI environments split into train, val-seen, and val-unseen sets with 16, 16, and 4 environments, respectively. Environments in val-seen split are the same as training, while the val-unseen split contains held-out environments. We collect 8000, 3200, and 800 episodes for the three splits, with an average trajectory length of approximately 10 m. The dataset includes diverse sign appearances and provides each target-relevant directional cue in two forms: a visual crop and a structured direction category, both paired with the cue bounding box.

Baselines: We compare START with three representative baselines in the form of visual crop as cue representation. The **Rule-based** baseline [25] follows a modular signage-guided navigation pipeline and is provided with ground-truth directional cues and local metric-topological information for subgoal selection. The **VLM-based** baseline uses Qwen3 [40] to reason navigation actions from the current observation and parsed sign cues; the exact prompt, action set, and decoding rules are provided in the supplementary material. The **ViNT** baseline [41] is adapted to **SignNav** by replacing its goal encoder with a signage-cue encoder and retrained under the same dataset split. These baselines cover rule-based planning, vision-language reasoning, and learning-based visual navigation.

Metrics: We report Success Rate (SR), normalized Dynamic Time Warping (NDTW), and Success weighted by normalized Dynamic Time Warping (SDTW) [1], [42]. SR measures whether the agent stops within 1.0 m of the target, while NDTW and SDTW evaluate trajectory similarity and successful path following. In addition, we report Correct Sign-Following Length

TABLE I
NAVIGATION PERFORMANCE COMPARISON WITH BASELINES

Methods	Val-Seen				Val-Unseen			
	SR	NDTW	SDTW	CSFL	SR	NDTW	SDTW	CSFL
VLM-based	0.35	0.39	0.27	0.50	0.42	0.47	0.34	0.53
Rule-based [25]	0.38	0.57	0.31	0.46	0.42	0.58	0.35	0.57
ViNT [41]	0.59	0.52	0.48	0.60	0.46	0.42	0.38	0.53
START (w/o DAgger)	0.65	0.59	0.52	0.67	0.49	0.47	0.38	0.57
START (w/ DAgger)	0.83	0.72	0.68	0.77	0.57	0.53	0.44	0.62

TABLE II
EFFECT OF CUE REPRESENTATION AND BOUNDING-BOX ABLATION

Cue Representation	BBox	Val-Unseen			
		SR	NDTW	SDTW	CSFL
Visual (w/o DAgger)	w/ bbox	0.49	0.47	0.38	0.57
	w/o bbox	0.43	0.42	0.34	0.52
Category (w/o DAgger)	w/ bbox	0.61	0.58	0.49	0.65
	w/o bbox	0.33	0.36	0.25	0.51
No semantic cue	w/ bbox	0.38	0.34	0.29	0.48

(CSFL), which measures the number of ground-truth signs correctly followed before the agent gets off the reference path. CSFL complements trajectory-level metrics by directly evaluating how long the agent remains consistent with the intended signage sequence.

Implementation details: For START, RGB and depth observations are encoded by lightweight CNN encoders, and the signage cue is encoded from the visual crop using ResNet-50 [37]. In the experiment of cue representation, the signage cue in the form of structured direction category is encoded using a learnable embedding. The Spatial-Aware Transformer adopts a ViT-B/32-style architecture [38], and the Temporal-Aware Transformer follows a BERT-like Transformer design [43]. Both modules use hidden size 768, 6 layers, and 6 attention heads. We first train the policy with teacher forcing and then fine-tune it using DAgger [39]. All simulation evaluations are conducted on one NVIDIA RTX 4090 GPU.

B. Main Results

Navigation performance: Table I compares START with representative baselines on the LSI-Dataset. START achieves the best overall performance on both val-seen and val-unseen splits. Compared with ViNT, START with DAgger improves SR from 0.59 to 0.83 on val-seen and from 0.46 to 0.57 on val-unseen, while also improving NDTW, SDTW, and CSFL. The rule-based baseline uses hand-crafted decisions and local topological information, but its performance remains limited by fixed heuristics and scene-dependent route structures. The VLM-based baseline can recognize directional cues in many cases, but its closed-loop low-level action prediction is unstable. ViNT provides a stronger learning-based baseline, yet lacks explicit cue-observation grounding and temporal memory. The improvement of START, especially in CSFL, indicates that the proposed spatial-temporal policy follows the intended signage sequence more consistently before deviating from the reference route. DAgger further improves closed-loop robustness by reducing the effect of covariate shift during execution.

Effect of cue representation: Table II studies different cue interfaces and the contribution of bounding-box information on the val-unseen split. The structured direction category with

TABLE III
QUANTITATIVE RESULTS OF REAL-WORLD DEPLOYMENT

Metrics	Lifts		Restroom		Pantry		Avg.
	1	2	3	4	5	6	
Success	1	1	1	0	1	1	5 / 6
Moved Distance (m)	26.1	25.8	15.3	23.4	15.7	15.3	20.3
Total Time (s)	212	201	133	173	121	104	157

TABLE IV
RUNTIME OF EACH MODULE IN THE DEPLOYMENT SYSTEM

Modules	Avg. Time (ms)	Avg. Frequency (Hz)
Transmission (RGB)	~280	3.57
Qwen3	~1800	0.55
DepthAnything-V2	~110	9.10
START Inference	~25	40.00
Transmission (Action)	~5	200.00
Total	2220	0.45

bounding box achieves the best performance, improving SR from 0.49 to 0.61 over the visual-crop cue with bounding box. This indicates that, when an upstream parser is available, representing the signage cue as a direction category can reduce dependence on low-level arrow appearance and improve generalization. Nevertheless, we use the visual crop as the default cue interface in the main experiments and subsequent ablation studies, since it preserves the original visual-cue grounding setting and directly evaluates whether START can learn from iconic signage observations. Removing the bounding box degrades both visual and category-based variants, showing that cue location is important for grounding the directional cue in the egocentric scene. The no-cue variant performs worse than the full cue-conditioned models, confirming that START benefits from both cue semantic and spatial cue localization rather than relying only on scene geometry.

Real-world deployment: We deploy START on a DeepRobotics Lite2 robot equipped with an Intel RealSense D435i camera in an office environment. At each step, the robot captures and transmits the RGB image to a remote desktop through wireless LAN. Qwen3-8B [40] is used as the upstream sign parser to extract the target-relevant directional cue and its crop location from the RGB observation, while DepthAnything-V2 [44] is used to generate a refined depth image for more stable geometric perception. START then takes the RGB image, generated depth image, parsed cue, and cue location as input, and predicts a discrete navigation action that is transmitted back to the robot for execution.

As shown in Table III, START succeeds in 5 out of 6 trials across three target types: lifts, restroom, and pantry. The robot travels 20.3 m on average and spends 157 s per trial. These quantitative trials complement the qualitative demonstration and show that the proposed downstream policy can use target-relevant signage cues in physical environments. Table IV reports the runtime of each deployment module. START inference runs at approximately 40 Hz, while the full system operates at about 0.45 Hz. The main latency comes from WLAN RGB transmission and Qwen3-8B signage parsing rather than from START itself, suggesting that faster onboard sign parsing and communication will be important for deployment in more dynamic public environments.

TABLE V
ABLATION STUDY OF SPATIAL- AND TEMPORAL-AWARE MODULES

Spatial-Aware	Temporal-Aware	Val-Unseen			
		SR	NDTW	SDTW	CSFL
×	✓	0.35	0.39	0.28	0.49
✓	×	0.39	0.41	0.32	0.48
✓	✓	0.49	0.47	0.38	0.57

TABLE VI
ABLATION STUDY UNDER DIFFERENT SIGNAGE-CUE DROPOUT RATIOS

Dropout Ratio	w/ Temporal-Aware				w/o Temporal-Aware			
	SR	NDTW	SDTW	CSFL	SR	NDTW	SDTW	CSFL
0.0	0.49	0.47	0.38	0.57	0.39	0.41	0.32	0.48
0.2	0.47	0.45	0.37	0.55	0.40	0.42	0.33	0.50
0.4	0.47	0.44	0.37	0.54	0.39	0.40	0.32	0.48
0.6	0.46	0.44	0.36	0.53	0.34	0.39	0.28	0.46
0.8	0.42	0.42	0.33	0.51	0.27	0.33	0.22	0.40

C. Ablation Studies

How important are the spatial- and temporal-aware modules? Table V reports the module ablation results on the val-unseen split. Removing either module degrades performance, while the full START model achieves the best results across all metrics. The spatial-only variant performs slightly better than the temporal-only variant in SR, NDTW, and SDTW, indicating that grounding the directional cue in the current egocentric observation is essential for action prediction. However, the full model further improves SR from 0.39 to 0.49 and CSFL from 0.48 to 0.57, showing that temporal reasoning helps the agent follow the intended signage sequence more consistently. These results support the complementary roles of spatial grounding and temporal memory in directional signage-cue guided navigation.

How does bounding-box position affect spatial grounding? We analyze the effect of bounding-box position using the ablation in Table II. Removing the bounding box consistently degrades performance for both cue interfaces. For the visual-crop cue, SR decreases from 0.49 to 0.43 and CSFL decreases from 0.57 to 0.52. For the structured category cue, the degradation is more pronounced, with SR dropping from 0.61 to 0.33 and SDTW from 0.49 to 0.25. These results indicate that the cue semantics alone are insufficient for robust navigation: the agent also needs the image-space cue location to ground the directional instruction in the current egocentric scene. The bounding box therefore provides important spatial information rather than merely serving as an auxiliary input.

How robust is START to sparse signage cues? Table VI evaluates the effect of temporal memory under different signage-cue dropout ratios. Specifically, we simulate different levels of signage sparsity by randomly dropping out the ground-truth signage observations along each episode with a certain probability. As the dropout ratio increases, the performance of both variants decreases, indicating that visible signage cues are important for accurate navigation. However, the model with the temporal-aware module consistently outperforms the variant without it across all dropout settings. The advantage becomes more evident under severe sparsity: at a dropout ratio of 0.8, the temporal-aware model improves SR from 0.27 to 0.42 and CSFL from 0.40 to 0.51. These results show that temporal memory helps START retain previously observed directional cues and maintain

TABLE VII
ABLATION STUDY OF DELAYED-TURN SETTINGS

Settings	Val-Unseen			
	SR	NDTW	SDTW	CSFL
Original	0.49	0.47	0.38	0.57
Delayed Turn (0.5 m)	0.49	0.47	0.38	0.57
Delayed Turn (1.0 m)	0.49	0.47	0.38	0.57
Delayed Turn (1.5 m)	0.50	0.47	0.39	0.57
Delayed Turn (2.0 m)	0.46	0.42	0.35	0.53

TABLE VIII
ABLATION STUDY OF DIFFERENT INPUT MODALITIES

RGB	Depth	Val-Unseen			
		SR	NDTW	SDTW	CSFL
✓	×	0.41	0.42	0.31	0.54
×	✓	0.56	0.52	0.45	0.62
✓	✓	0.49	0.47	0.38	0.57

the intended signage-following behavior when current cues are missing.

Can START handle delayed turning after observing signage? Table VII evaluates whether START can execute a turn after the signage cue has been observed earlier than the actual junction. In this setting, the current signage cue is removed within a specified distance before the junction, forcing the agent to rely on its previous cue observation and current scene geometry. START maintains nearly unchanged performance for delayed distances from 0.5 m to 1.5 m, with SR around 0.49 ~ 0.50 and CSFL remaining at 0.57. Performance decreases at 2.0 m, where the cue must be retained over a longer interval and larger trajectory uncertainty may accumulate. These results indicate that START can handle moderate delayed-turn cases through temporal memory and spatial grounding, while longer cue-action delays remain challenging.

What is the impact of different input modalities? Table VIII reports the effect of RGB and depth observations on START. The depth-only variant achieves the best performance, improving SR from 0.49 to 0.56 and CSFL from 0.57 to 0.62 compared with RGB-D input. This suggests that geometry provides strong cues for local traversability and route structure in the current LSI-Dataset. The RGB-only variant performs worse, indicating that appearance alone is less reliable for closed-loop control in these indoor scenes. Nevertheless, RGB remains necessary in the complete real-world system for upstream signage parsing and cue extraction. These results also highlight a limitation of the current benchmark: the policy may rely heavily on geometric structure, and future work should incorporate richer semantic supervision and more visually diverse environments where geometry alone is insufficient.

VI. CONCLUSION AND LIMITATIONS

This paper presented **SignNav**, a directional signage-cue guided visual navigation task for LSI environments. We constructed the **LSI-Dataset** with procedurally injected signage, diverse sign appearances, and two cue representations: visual cue crop and structured direction category. We also proposed **START**, a Spatial-Temporal Aware Transformer that grounds target-relevant directional cues in egocentric observations and uses temporal memory to handle sparse or delayed signage. Experiments on the LSI-Dataset, cue-representation ablations,

sparse-cue evaluations, and real-world robot trials demonstrate the effectiveness of START for downstream signage-cue conditioned navigation.

While these results show the promise of directional signage cues for LSI navigation, they also point to several limitations of the current formulation. This work focuses on parsed directional cues rather than full signage understanding, such as OCR, compound sign parsing, or open-ended visual sign grounding. The current cue and action spaces cover only basic directional behaviors and do not model richer public-space instructions such as stairs, escalators, or multi-step compound signs. The benchmark also assumes mostly static environments and does not address dynamic crowds or robust backtracking after wrong turns. Future work will extend SignNav toward richer signage semantics, more complex navigation behaviors, and stronger recovery mechanisms.

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